

Lesson 11

Modular Arithmetic

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11.1 Introduction

We already know that divisibility is a fundamental concept of number theory. In this lesson, again we study the divisibility in different point of view, with a rich algebraic structure. Now we approach the divisibility through the arithmetic of remainders. The theory of congruence is the commonly known name for this approach. The basic idea of congruence is to perform arithmetic with the remainders obtained upon division. The advantage of this approach is, the some information regarding the divisibility is contained from divisions. The arithmetic of congruence is generally called as the Modular Arithmetic.

11.2 Basic properties of congruences

11.2.1 Definition

Let n be a fixed positive integer. Two integers a and b are said to be congruent modulo n , symbolized by $a \equiv b \pmod{n}$ if n divides the difference $a - b$; that is, provided that $a - b = kn$ for some integer k .

11.2.2 Example: Show that $-15 \equiv -64 \pmod{7}$, $101 \equiv 2 \pmod{11}$ and $11 \equiv -4 \pmod{5}$

Answer:

Observe that $-15 - (-64) = 49$ therefore $7 | (-15 - (-64))$ and hence $-15 \equiv -64 \pmod{7}$.

$101 - 2 = 99$, therefore $11 | (101 - 2)$ and hence $101 \equiv 2 \pmod{11}$.

$11 - (-4) = 15$, therefore $5 | (11 - (-4))$ and hence $11 \equiv -4 \pmod{5}$.

Note:

When $n \nmid (a-b)$, we say that “ a is incongruent to b modulo n ” (or “ a is not congruent to b modulo n ”), and in this case we write $a \not\equiv b \pmod{n}$.

For example $15 \not\equiv 12 \pmod{7}$, since 7 fails to divide $15 - 12 = 3$.

11.2.3 Theorem

For any integers a and b , $a \equiv b \pmod{n}$ if and only if a and b leave the same nonnegative remainder when divided by n .

Proof:

First suppose $a \equiv b \pmod{n}$; so that $a = b + kn$ for some integer k .

Upon division by n , b leaves a certain remainder, say r ; i.e. $b = qn + r$ where $0 \leq r < n$.

Therefore $a = b + kn = qn + r + kn = (q + k)n + r$. Which indicates that a has the same remainder as b .

Conversely suppose a and b have the same remainder upon division by n . Let it be r .

Hence we can write $a = q_1n + r$ and $b = q_2n + r$, where $0 \leq r < n$.

Then $a - b = (q_1n + r) - (q_2n + r) = (q_1 - q_2)n$.

Hence $n | (a - b)$ i.e. $a \equiv b \pmod{n}$. ■

11.2.4 Example

Since the integers -56 and -11 can be expressed in the form

$$-56 = (-7)9 + 7 \text{ and } -11 = (-2)9 + 7$$

With the same remainder 7, above theorem, tells us that $-56 \equiv -11 \pmod{9}$.

On the other direction, $-31 \equiv 11 \pmod{7}$ implies that -31 and 11 have the same remainder when divided by 7;

Also by checking this is true, since $-31 = (-5)7 + 4$, $11 = 1 \cdot 7 + 4$.

From the next theorem we prove the basic algebraic properties of the congruence.

11.2.5 Theorem

Let $n > 1$ be fixed and a, b, c, d be arbitrary integers. Then the following properties hold:

- (a) $a \equiv a \pmod{n}$
- (b) If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$
- (c) If $a \equiv b \pmod{n}$, and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.
- (d) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$ and $ac \equiv bd \pmod{n}$.
- (e) If $a \equiv b \pmod{n}$, then $a + c \equiv b + c \pmod{n}$ and $ac \equiv bc \pmod{n}$.
- (f) If $a \equiv b \pmod{n}$, then $a^k \equiv b^k \pmod{n}$ for any positive integer k .

Proof:

- (a) For any integer a , we have $a - a = 0$, so $a \equiv a \pmod{n}$.
- (b) Suppose $a \equiv b \pmod{n}$. Then $a - b = kn$ for some integer k . Hence $b - a = (-k)n$ and since $-k$ is an integer, we get $b \equiv a \pmod{n}$.
- (c) Suppose $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. Then there exist integers h and k satisfying $a - b = hn$ and $b - c = kn$. It follows that $a - c = (a - b) + (b - c) = hn + kn = (h + k)n$ which is $a \equiv c \pmod{n}$ in congruence notation.
- (d) Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$.

Then $a = b = k_1n$ and $c - d = k_2n$ for some integers k_1 and k_2 . adding these equations, we obtain

$$(a + c) - (b + d) = (a - b) + (c - d) = k_1n + k_2n = (k_1 + k_2)n$$

i.e, $(a + c) \equiv (b + d) \pmod{n}$.

$$\text{Now, } ac = (b + k_1n)(d + k_2n) = bd + (bk_2 + dk_1 + k_1k_2n)n$$

Since $(bk_2 + dk_1 + k_1k_2n)$ is an integer, this says that $ac - bd$ is divisible by n , hence $ac \equiv bd \pmod{n}$.

(e) Suppose $a \equiv b \pmod{n}$. Since $c \equiv c \pmod{n}$. (by part (a))

We get, $a + c \equiv b + c \pmod{n}$ and $ac \equiv bc \pmod{n}$. (by part (d))

(f) The result is true for $k = 1$ by hypothesis. We will assume the result is true for some positive integer k . Now since $a \equiv b \pmod{n}$ and $a^k \equiv b^k \pmod{n}$, by part (d) we get $aa^k \equiv bb^k \pmod{n}$. i.e. $a^{k+1} \equiv b^{k+1} \pmod{n}$. Hence the result.

Notation:

Let n be fixed positive integer. Let a be any integer. Then according to the division algorithm there exist unique integers q and r satisfying $a = qn + r$ with $0 \leq r < n$. the notation $a \pmod{n}$ represent that unique remainder r . i.e $a \pmod{n} = r$.

6.2.6 Example

$$\begin{aligned} 371 \pmod{10} &= 1 \\ 71 \pmod{13} &= 6 \\ -23 \pmod{10} &= 7 \\ 91 \pmod{13} &= 0 \end{aligned}$$

11.2.7 Example

Without evaluating 2^{32} determine $2^{32} \pmod{17}$.

Answer:

The computation can be done using the properties of congruences.

$$2^3 \equiv 8 \pmod{17}, \quad 2^4 \equiv 16 \pmod{17}, \quad 2^5 \equiv 32 \equiv 15 \pmod{17}, \quad 2^{10} \equiv 15^2 \equiv 4 \pmod{17},$$

$$2^{30} \equiv 4^3 \equiv 64 \equiv 13 \pmod{17}, \quad 2^{32} \equiv 13 \cdot 4 \equiv 52 \equiv 1 \pmod{17},$$

Hence $2^{32} \pmod{17} = 1$.

2nd method

$$2^4 \equiv 16 \equiv -1 \pmod{17}, \quad 2^8 \equiv (-1)^2 \equiv 1 \pmod{17}, \quad 2^{16} \equiv 1 \pmod{17}, \quad 2^{32} \equiv 1 \pmod{17}$$

Hence $2^{32} \pmod{17} = 1$

Activity 01

(i) Prove that:

(a) If $a \equiv b \pmod{n}$ and $m|n$, then $a \equiv b \pmod{m}$

(b) If $a \equiv b \pmod{n}$ and $c > 0$, then $ca \equiv cb \pmod{n}$

(ii) Prove that If $a \equiv b \pmod{n_1}$ and $a \equiv c \pmod{n_2}$,

then $b \equiv c \pmod{n}$, where the integer $n = \gcd(n_1, n_2)$

(iii) Give an example to show that $a^2 \equiv b^2 \pmod{n}$ need not imply that $a \equiv b \pmod{n}$
